

GUFAN WANG

Robotics Engineering Undergraduate | Real-robot locomotion, Sim-to-Real, and real-time systems

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Jiaxing, Zhejiang, China
Expected graduation: June 2027

Education

University of Science and Technology Beijing (USTB)

B.Eng. in Robotics Engineering; GPA: 3.5/4.0

Beijing, China
Sep 2023 - Jun 2027

Relevant coursework: Robotics, Automatic Control Theory, Machine Learning, Reinforcement Learning and Optimal Control, Deep Learning, Mechanical Design.

Research Interests

Reliable legged and wheel-legged locomotion; Sim-to-Real transfer; timing-aware state estimation; real-time optimization and control under sensing, actuation, and model uncertainty.

Research Experience

25 kg Wheel-Legged Robot - RL Locomotion and Real-Robot Deployment

May 2026 - Aug 2026

USTB Advanced Robotics Laboratory

- Built a multi-rate locomotion path from PPO policy inference to 16 actuators through ONNX, Linux shared memory, dual STM32F407 MCUs, and four CAN buses.
- Trained and deployed standing, wheeled locomotion, stair behavior, and disturbance-recovery policies using Isaac Lab / RSL-RL; exported the policy to ONNX with an explicit 57-D observation and 16-D action contract.
- Investigated disturbance-induced oscillation by broadening dynamics randomization and retaining synchronization, feedback-freshness, and watchdog checks in the deployment loop.

55 kg Quadruped - PPO Locomotion and EtherCAT Sim-to-Real

Sep 2025 - Jun 2026

USTB Advanced Robotics Laboratory; joint project with Tsinghua University and the Chinese PLA General Hospital

- Developed the path from a 45-D PPO observation to 12 joint targets, implemented at 50 Hz, with 1 kHz PD control and 2 kHz SOEM EtherCAT PDO exchange.
- Integrated 12 CiA-402 Profile-Torque drives, IMU and joint feedback, action mapping, state-machine logic, and shared-memory interfaces for the hardware control stack.
- Recorded desired and measured joint trajectories to diagnose model error, actuator delay, static friction, and interface mismatch during hardware deployment.

Technical Skills

Learning and simulation: PPO, Isaac Lab, RSL-RL, MuJoCo, Sim-to-Real, dynamics randomization.

Robotics and systems: C++, Python, C, Linux, ROS 2, EtherCAT, CAN, PD control, shared memory, embedded MCU development.

Machine learning and vision: PyTorch, PINNs, Transformer/MLP regression, YOLO-based detection.

Additional Robotics and Field Experience

20-Motor Upper-Limb / Exoskeleton System - Ubuntu, ROS 2, and EtherCAT Migration

Sep 2024 - Jan 2025

USTB Advanced Robotics Laboratory

- Migrated a Windows-oriented 20-motor control workflow toward Ubuntu and ROS 2; adapted EtherCAT master/slave communication and motor-control interfaces.
- Integrated a control card without native Linux support and connected data-collection and command paths to the ROS 2 system.

Magnetic Wall-Climbing Inspection Robot - Vision and Field Deployment

Sep 2024 - Sep 2025

USTB Advanced Robotics Laboratory; Wudongde Hydropower Station field deployment

- Contributed to metal-surface defect data collection, model comparison, and YOLOv11 deployment for a magnetic climbing inspection robot.
- Participated in field work under dust, vibration, illumination variation, and viewpoint constraints.

Scientific Machine Learning Experience

Time-Varying PINN for Furnace Temperature Prediction

Jun 2025 - Dec 2025

USTB Automation and Control Laboratory

- Extended a static PINN workflow to full-cycle temperature-field prediction with nonuniform heat-source conditions; combined a 2D finite-difference reference path with PDE and boundary-condition losses.

Data-Driven Magnesium Alloy Property Modeling

Jun 2025 - Sep 2025

Taiyuan University of Technology Forming Laboratory

- Worked on data augmentation and regression workflows using Transformer, MLP, and stacking models for alloy-property prediction from composition and process variables.

Research Outputs

- PINN-Based Material Temperature Forecasting and Heating-Furnace Curve Optimization - second author; accepted manuscript.
- Efficient Design of Magnesium Alloys with Superior Mechanical Properties Using Deep Learning - second author; manuscript under review.
- Two utility model patents - one sole inventor and one co-inventor, according to the current CV.

Honors and Awards

- National First Prize, China International College Innovation Competition, 2025.
- Beijing Grand Prize, Challenge Cup AI Track, 2025; National Third Prize, Challenge Cup, 2025.
- USTB Freshman Scholarship and People's Scholarship (Third Class).